

Oral Defense Script: Anatomy-Aware Pose Estimation on Lightweight Networks

Presenter: Flavio Massaroni (Slides 6-10) **Target Duration:** ~4.5 to 5 Minutes (Approx. 680 words at 140 WPM) **Tone:** Rigorous, academic, data-driven, and defensive.

Slide 6: Result 1 — Severity Weighting

[Time Marker: 0:00 – 1:00 | Word Count: ~145]

"Moving into the experimental phase, I will begin by presenting our first core architectural intervention: the **Severity Weighting** mechanism.

Our initial failure analysis revealed a phenomenon we term **gradient dilution**. Poses exhibiting severe anatomical violations constitute only about 17% of any given training batch. Under a standard batch-mean loss formulation, the learning signal generated by these critical violations is severely diluted by the remaining 83% of compliant or near-compliant samples — the network never receives sufficient gradient pressure to correct its worst anatomical errors.

To resolve this, we introduced a dynamic severity-weighted penalty factor, parameterized by $\alpha = 2.0$, applied to the excess violation in the bone ratio term: the penalty scales as $(1 + \alpha \cdot \text{excess})$, amplifying the gradient precisely where violations are most severe. This is independent from our coordinate extraction pipeline, which uses a fixed soft-argmax temperature of $\beta = 50$ — calibrated separately to minimize the localization gap between training and evaluation.

As shown in the table, this targeted gradient reinforcement successfully drove the COCO AVR down from 0.3062 to 0.2968, and the zero-shot OCHuman AVR from 0.4822 to 0.4775, while keeping the model's Average Precision completely intact. Crucially, to ensure these drops represent real structural optimization rather than stochastic drift, our entire pipeline enforces absolute determinism with fixed seeding, zero-worker data loaders, and deterministic cuDNN flag configurations, clearing the inherent noise floor."

Slide 7: The Insight — The Foreshortening Floor

[Time Marker: 1:00 – 1:45 | Word Count: ~115]

"Despite the success of the Severity Weighting loss, we encountered a persistent performance barrier. This led to our central diagnostic insight: the **foreshortening floor**.

A granular diagnostic breakdown of our failures demonstrated that **63.3% of all severe bone ratio violations** are physically consistent with true 2D perspective foreshortening — specifically, where a limb is projected directly along the camera's optical axis, causing its apparent 2D length to drop below 25% of the torso length.

This represents an irreducible structural floor, not an algorithmic flaw. In a purely 2D representation space, there is an absolute informational ambiguity: a 2D metric cannot distinguish an anatomically incorrect prediction from a physically valid, foreshortened limb. Forcing the loss to penalize these cases would be counterproductive — it would compel the network to artificially extend geometrically valid limbs in 2D projection, actively worsening the very metric we are trying to improve."

Slide 8: Solution — Foreshortening Augmentation

[Time Marker: 1:45 - 2:30 | Word Count: ~135]

"To address this physical limitation without compromising the loss landscape, we shifted our strategy to the data side by introducing a dedicated **Foreshortening Augmentation** pipeline during training.

Instead of fighting the geometric floor with the loss function, we explicitly exposed our MobileNetV3 backbone to oblique spatial perspectives. We implemented a synthetic **Perspective Warp**, applying homography matrices defined as $H = K \cdot R \cdot K^{-1}$, which simulates realistic camera pitch and yaw tilts up to $\pm 12^\circ$. The synthetic focal length is set equal to the canvas width — a standard approximation for tightly cropped person detections.

Critically, this operation maintains mathematical coherence: both the image canvas and the Ground Truth keypoints are warped simultaneously using the exact same homographic transformation, ensuring zero spatial misalignment. Our geometric analysis confirmed a negligible ground-truth conflict rate of only **0.87%** at this tilt range. Finally, this augmentation acts as a strict complement to our Skeletal Topology Loss — the loss continues to operate unchanged — and is deployed exclusively in the training phase, leaving our validation protocol pristine."

Slide 9: Final Results

[Time Marker: 2:30 - 3:45 | Word Count: ~170]

"This brings us to our complete three-arm comparative analysis. When evaluating our final model — designated as **sev_aug_E6** — the synergy between our loss-side methodology and data-side augmentation becomes clear.

The aggregate AVR drops consistently across both datasets, with our final model outperforming both the baseline and the standalone severity model globally. The most compelling validation occurs on the **zero-shot OCHuman dataset** — completely unseen during training, characterized by extreme occlusions and complex poses. Here, the overall AVR drops by 2.52 points relative to the baseline — a **5.2% relative improvement on a dataset the model has never seen**, and more than twice our documented stochastic noise floor.

Our primary headline result lies within the specific `(bone_ratio)` sub-category on OCHuman. The metric drops from 0.5495 down to 0.5258. This delta of **-0.0238** stands approximately **2.4 times above our noise floor**, confirming a robust, statistically significant signal — not a fluctuation.

Most importantly, we achieve this anatomical compliance at **zero cost to precision**. Average Precision remains consistent across all configurations, with OCHuman AP even recording a slight technical increase of +0.0001. The robustness price is minimal, manifesting solely as prediction variance on simple, non-occluded poses — without degrading mean performance."

Slide 10: Conclusion and Future Works

[Time Marker: 3:45 - 4:45 | Word Count: ~115]

"In conclusion, this work demonstrates that an ultra-lightweight, 2.5-million-parameter network can successfully internalize explicit biomechanical priors derived from the Winter and AAOS literature. By aligning our Skeletal Topology Loss 1:1 with our Anatomical Violation Rate metric, the model achieves meaningful zero-shot generalization under severe real-world occlusions — with no trade-off in accuracy.

To address our current limitations, our future roadmap follows three distinct tracks: First, overcoming the foreshortening floor by implementing an offline, lightweight **Depth Head** to explicitly resolve the 2D geometric ambiguity that our metric cannot disambiguate. Second, completing our ablation matrix with a standalone *Baseline plus Augmentation* arm, to formally isolate the contribution of the STL-augmentation interaction from augmentation alone. Third, incorporating **structured occlusion masks** in the style of Han et al., to compel the model to infer missing joints entirely from global pose context — a complementary data-side strategy to our perspective warp.

Ultimately, by ensuring anatomical validity on lightweight architectures, we bridge the gap between theoretical computer vision and safe, real-time edge deployment for Human-Robot Interaction. Thank you — I am open to your questions."